



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Object Oriented Programming (Spring-2024)

LAB No. 3

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Objective of Lab No. 1:

After performing lab3, students will be able to:

- Implement Logical and Relational Operators in Java
- Implement Selection Statements in Java
- Implement Iteration Statements in Java
- Implement jump statements in java
- Implement Nested Loops in java
- Use All the above concepts to solve real life problems

Recap of Last Lab:

Objective-01: Create and Run First Java program (“Hello World”)

Data Types in Java

Data types classify the different values to be stored in the variable. In java, there are two types of data types:

1. Primitive Data Types
2. Non-primitive Data Types

Primitive Data Types

Primitive Data Types are predefined and available within the Java language. Primitive values do not share state with other primitive values.

There are 8 primitive types: byte, short, int, long, char, float, double, and boolean

Integer data types



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```
byte (1 byte)
short (2 bytes)
int (4 bytes)
long (8 bytes)
```

Floating Data Type

```
float (4 bytes)
double (8 bytes)
```

Textual Data Type

```
char (2 bytes)
```

Logical

```
boolean (1 bit) (true/false)
```

Input & Output

Output in Java Syntax:

```
System.out.println("Hello World");
```

Input in Java Syntax:

```
// import library
```

```
import java.util.Scanner;
```

```
// Creating scanner object
```

```
Scanner obj = new Scanner(System.in);
```

```
Taking input from user
```

```
int num = obj.nextInt(); // Integer Input
```

```
double num = obj.nextDouble(); // Double Input
```

Java Variable Type Conversion & Type Casting

A variable of one type can receive the value of another type. Here there are 2 cases.

Case 1) Variable of smaller capacity is be assigned to another variable of bigger capacity.



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```
double d ;  
int i = 10;  
d = i;
```

This process is Automatic, and non-explicit is known as **Conversion**

Case 2) Variable of larger capacity is be assigned to another variable of smaller capacity.

```
double d = 10;  
int i;  
i = (int) d
```

Type Cast
Operator

In such cases, you have to explicitly specify the **type cast operator**. **This process is known as Type Casting.**

In case, you do not specify a type cast operator; the compiler gives an error. Since this rule is enforced by the compiler, it makes the programmer aware that the conversion he is about to do may cause some loss in data and prevents **accidental losses**.

Example: To Understand Type Casting

```
class Demo {  
    public static void main(String args[]) {  
        byte x;  
        int a = 270;  
        double b = 128.128;  
        System.out.println("int converted to byte");  
        x = (byte) a;  
        System.out.println("a and x " + a + " " + x);  
        System.out.println("double converted to int");  
    }  
}
```



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```
a = (int) b;  
System.out.println("b and a " + b + " " + a);  
System.out.println("\ndouble converted to byte");  
x = (byte)b;  
System.out.println("b and x " + b + " " + x);  
}  
}
```

Output:

```
int converted to byte  
a and x 270 14  
double converted to int  
b and a 128.128 128  
double converted to byte  
b and x 128.128 -128
```



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Practice:

Task 1: Arithmetic Operators

Write a Java program that:

- Takes two numbers (int and double) as input
- Performs **addition, subtraction, multiplication, division**
- Displays the result of each operation

Task 2: Unary Operator

Write a Java program to:

- Declare an integer variable
- Apply **unary increment (++) and decrement (--) operators**
- Print the value before and after each operation

Task 3: Modulus Operator

Write a Java program that:

- Takes two integers as input
- Displays the **remainder using % operator**
- Test the case where the **left operand is smaller than the right operand**

Task 4: Floating-Point Modulus

Write a Java program that:

- Takes a **floating-point number as input**
- Finds and displays the **remainder when divided by 5**



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Task 5: Compound Assignment Operators

Write a Java program that:

- Initializes an integer variable
- Applies the following compound operators:
 - +=
 - -=
 - *=
 - /=
- Displays the value after each operation

Task 6: Bitwise AND, OR, XOR

Write a Java program that:

- Declares two integers
- Performs **bitwise AND (&)**, **OR (|)**, and **XOR (^)**
- Displays results in **decimal form**

Task 7: Bitwise NOT Operator

Write a Java program that:

- Takes an integer as input
- Applies **bitwise NOT (~)**
- Displays the original and resulting value

Task 8: Binary Shift Operators

Write a Java program that:

- Takes an integer as input
- Applies:
 - **Left shift (<<)**
 - **Right shift (>>)**
- Display results and verify:
 - Left shift $\approx$ multiply by 2



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- Right shift $\approx$ divide by 2

Task 9: Relational Operators

Write a Java program that:

- Takes two integers as input
- Uses **relational operators** (>, <, >=, <=, ==, !=)
- Prints the boolean result of each comparison

Task 10: Logical vs Short-Circuit AND

Write a Java program that:

- Demonstrates the difference between:
 - & (logical AND)
 - && (short-circuit AND)
- Use conditions where the second expression should not be evaluated

Task 11: Ternary Operator (From Lecture Task)

Write a Java program that:

- Takes **salary as input**
- Uses **only the ternary operator** to:
 - Print "Managerial Level" if salary > 70000
 - Otherwise print "Staff Level"



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Lab Exercises:

1. Write a Java program that:
 - i. Takes **basic salary** and **years of experience** as input.
 - ii. Uses **arithmetic and compound assignment operators** to calculate:
 - iii. Allowance (+10% if experience ≥ 5 years)
 - iv. Uses **relational and logical operators** to check:
 - v. If final salary > 70000 and experience ≥ 5
 - vi. Uses the **ternary operator** to print:
 - vii. "Senior Manager" OR "Junior Staff"
2. Bitwise Access Control System - Design a Java program that:
 - i. Stores user permissions using an **integer (bit flags)**.
 - ii. Uses **bitwise OR ($|=$)** to grant permissions.
 - iii. Uses **bitwise AND ($\&$)** to check permissions.
 - iv. Uses **bitwise XOR ($\wedge=$)** to toggle permissions.
 - v. Displays the permission status after each operation.
3. Smart Calculator with Operator Selection - Write a Java program that:
 - i. Takes two numbers and an operator (+, -, *, /, %) from the user.
 - ii. Uses:
 - a. **Relational operators** to validate division by zero
 - b. **Arithmetic operators** to perform calculations
 - iii. Uses **ternary operator** to print:
 - a. "Valid Operation" OR "Invalid Operation"
 - iv. Prints the final result only if the operation is valid.
4. Binary Shift Performance Calculator - Write a Java program that:
 - i. Takes an integer input.
 - ii. Uses:
 - a. **Left shift ($<<$)** to multiply by 4
 - b. **Right shift ($>>$)** to divide by 2
 - iii. Uses **arithmetic operators** to verify the results.



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- iv. Uses **relational operators** to compare both results and display which is larger.
- 5. Student Result Classification System - Create a Java program that:
 - i. Takes marks of **3 subjects**.
 - ii. Calculates **total and average** using arithmetic operators.
 - iii. Uses **logical operators (&&, ||)** to:
 - a. Determine pass/fail (each subject ≥ 40)
 - iv. Uses **ternary operator** to assign grade:
 - a. "Distinction", "Pass", or "Fail"